MIL-A-9482(AV)
AMENDMENT 2
2 April 1981

Superseding
AMENDMENT 1
10 June 1955

MILITARY SPECIFICATION

ANTI-ICING EQUIPMENT FOR AIRCRAFT,

HEATED SURFACE TYPE, GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-A-9482, dated 2 September 1954.

PAGE 1

2.1, under SPECIFICATIONS, Military:

Delete:	MIL-E-7563	Electrical Equipment, Aircraft, Installation of, General Specification for
	MIL-D-9402	Design of Electrical Equipment in Aircraft, General Specification for
Add:	MIL-E-25499	Electrical Systems, Aircraft, Design and Installation of, General Specification for

Add boxed paragraph to bottom center of page.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Research and Development Command, ATTN: DRDAV-ERS, 4300 Goodfellow Boulevard, St. Louis, MO 63120, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1650

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:57Z
Check the source to verify that this is the current version before use.

PAGE 2

3.2.1 Delete and substitute: "3.2.1 All portions of the wings and stabilizing surfaces which are exposed to ice accumulations during flight, which would cause perceptible performance deterioration under the conditions specified in 3.3.1 and 3.3.2 shall be protected by the anti-icing system."

3.2.4, line 4: Delete "unrestricted" and substitute "required."

PAGE 3

3.4.4 Delete and substitute: "3.4.4 Water evaporation shall be determined from an integration of point evaporation rates taken at a minimum of three points on top and bottom surface, at a selected station, and shall not be based on average skin temperatures. Normally two of the points shall be in the impingement area and the third in the partially wetted area."

PAGE 5

3.6.3 Delete and substitute: "3.6.3 Distribution ducts shall be insulated as necessary to reduce temperature losses to values consistent with system requirements and to prevent damage to adjacent structure, wiring, or equipment. Figure 2 shows suggested practice in use of insulation."

3.6.6 Delete and substitute: "3.6.6 All ducts and connections used shall be structurally adequate to resist without rupture internal pressures positive and negative, up to two and one-half times the maximum design pressure at the design temperature. They shall not be subject to permanent deformation when exposed to 1.33 times the design pressure at the design temperature. The design pressure and temperature shall be that combination of pressure and temperature consistent with operating conditions which produces the maximum ratio of applied stress to allowable stress. Ducts which carry internal pressures in excess of 50 psig shall be capable of resisting 10,000 repeated cycles of pressure. One thousand of the cycles should be from 10 percent of design pressure to the design operating pressure at design temperature and the remaining cycles from 10 percent of design pressure to 90 percent at the design pressure at the design temperature. Ducts subjected to bending due to differential expansion of the duct system relative to its supporting structure shall be capable of resisting 10,000 deflection cycles, 10 percent of which shall be the maximum expected in the airplane and 90 percent of which shall be at 90 percent maximum deflection applied simultaneously and in phase with the internal pressure cycles. Rigorous analytical determination of stresses and comparison with established fatigue stress limits will be accepted as evidence of ability to withstand repeated cycles of bending and pressure for straight or slightly curved ducts of circular cross section. For other ducts this resistance must be demonstrated by test, under the actual or equivalent conditions."

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:57Z
Check the source to verify that this is the current version before use.

PAGE 6

Figure 2, Title. Delete and substitute: "Suggested Insulation Practices For Spanwise Flow Passages." Items (a) and (d): Delete "required" and substitute "suggested."

PAGE 7

3.6.14 Delete and substitute: "3.6.14 Duct supports shall be designed to minimize heat loss from the duct through the support and shall be designed structurally to load factors consistent with those applicable to the major aircraft structure. In exposed locations, where the supports may be used at steps or hand grips or where mechanics working in the area may lean against the support, the support shall be designed to withstand at least a 200 pound vertical load and 100 pound horizontal load. Where bellows are introduced to allow relative movement between two rigid ducts, the rigid ducts shall be supported immediately adjacent to the junction between the flexible and rigid sections. At critical locations the supports shall be adjustable to compensate for variations in dimensions within allowable manufacturing tolerances."

3.6.17 Delete MIL-D-9402 and MIL-E-7563 and substitute MIL-E-25499.

PAGE 8

Add the following new paragraph:

"3.6.20 Bellows or other types of flexible sections shall be capable of withstanding the same internal pressures without deformation or rupture and the same internal pressure cycles as the ducts they connect. In addition, they shall be capable of withstanding a number of axial, angular, and torsional displacements equivalent to 1,000 hours of aircraft flight time. If the internal flow Mach number exceeds 0.3 and the internal surface exposed to flow is a convoluted surface, the ability to withstand 100 hours of internal flow at maximum internal Mach number shall be demonstrated."

4.1.1, first line: Between "equipment" and "used" insert the words "including ducts."

PAGE 10

Add the following new paragraph:

6.3 Reclaimed Materials. The use of reclaimed materials shall be used to the maximum extent possible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:57Z
Check the source to verify that this is the current version before use.